

Worked examples for Lecture 21: Equilibrium figures

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These examples accompany Lecture 21. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory. Throughout, $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, the mean radius of the Earth is $b = 6371 \text{ km}$, its mass is $M = 5.972 \times 10^{24} \text{ kg}$, its mean density is $\bar{\rho} = 5515 \text{ kg m}^{-3}$, and its angular velocity is $\Omega = 7.292 \times 10^{-5} \text{ s}^{-1}$. Values quoted for the Preliminary Reference Earth Model (PREM) are from Dziewonski & Anderson (1981).

Example 1: Pressure inside a homogeneous non-rotating planet. A non-rotating planet of radius b has uniform density ρ . Show that the hydrostatic pressure is

$$p(r) = \frac{2}{3} \pi G \rho^2 (b^2 - r^2),$$

and express the central pressure in terms of the mass M and the surface gravity $g(b)$. Evaluate the central pressure for a planet with the mean radius and mean density of the Earth, compare with the value $3.64 \times 10^{11} \text{ Pa}$ at the centre of PREM, and explain the discrepancy.

Solution. For a spherically symmetric non-rotating planet the hydrostatic equilibrium equation reduces to

$$\frac{dp}{dr} + \rho g_0 = 0, \quad g_0(r) = \frac{d\phi_0}{dr} = \frac{GM(r)}{r^2}, \quad (1)$$

where $M(r)$ is the mass within radius r , and $g_0 > 0$ is the magnitude of the gravitational acceleration. The expression for g_0 follows from Gauss's law applied to Poisson's equation, or equivalently from Newton's shell theorem. For uniform density $M(r) = \frac{4}{3} \pi \rho r^3$, and so

$$g_0(r) = \frac{4}{3} \pi G \rho r, \quad (2)$$

which increases linearly from the centre to the surface. Integrating eq. (1) inwards from the surface, where $p(b) = 0$,

$$p(r) = \int_r^b \rho g_0(r') dr' = \frac{4}{3} \pi G \rho^2 \int_r^b r' dr' = \frac{2}{3} \pi G \rho^2 (b^2 - r^2), \quad (3)$$

as required. The pressure is a parabola in r , and its central value is

$$p(0) = \frac{2}{3} \pi G \rho^2 b^2. \quad (4)$$

Substituting $\rho = 3M/(4\pi b^3)$ we can write this as

$$p(0) = \frac{3GM^2}{8\pi b^4} = \frac{3g(b)^2}{8\pi G}, \quad (5)$$

where $g(b) = GM/b^2$ is the surface gravity. The second form is worth remembering: it says that the central pressure of a homogeneous planet is fixed by its surface gravity alone, and does not otherwise depend on its size.

For the Earth's mean density and radius, eq. (4) gives

$$p(0) = \frac{2}{3}\pi \times 6.674 \times 10^{-11} \times (5515)^2 \times (6.371 \times 10^6)^2 \text{ Pa} = 1.73 \times 10^{11} \text{ Pa}, \quad (6)$$

which is less than half the PREM value of $3.64 \times 10^{11} \text{ Pa}$. The discrepancy arises because the Earth's mass is concentrated towards its centre: the density in the core is roughly twice that in the mantle, and within each region density increases with depth through self-compression. Two effects then act in the same direction. First, concentrating mass towards the centre increases $M(r)/r^2$ at depth, so that g_0 within the Earth is everywhere larger than the linear profile of eq. (2); in PREM, g_0 is roughly constant at about 10 m s^{-2} through the whole mantle, and only falls towards zero within the core. Second, the weight per unit volume ρg_0 at depth is further increased by the larger density there. Both effects steepen the pressure gradient at depth, and hence raise the central pressure. The homogeneous model gets the mass and surface gravity right, but the pressure within a planet depends on how that mass is distributed with radius, and a homogeneous model places too much of it near the surface. The next example quantifies this with the simplest model that has a dense core.

Example 2: Two-layer planet. A non-rotating planet consists of a core of uniform density ρ_c and radius c , surrounded by a mantle of uniform density ρ_m extending to the surface $r = b$. Obtain the gravitational acceleration $g_0(r)$ and the hydrostatic pressure $p(r)$ in both layers. Taking $\rho_c = 11 \times 10^3 \text{ kg m}^{-3}$, $\rho_m = 4.5 \times 10^3 \text{ kg m}^{-3}$, $c = 3480 \text{ km}$, and $b = 6371 \text{ km}$, compute the mass, the moment of inertia factor $I/(Mb^2)$, the pressure at the core–mantle boundary (CMB), and the central pressure. Compare with the Earth's values $I/(Mb^2) = 0.3307$, and with PREM's $p(\text{CMB}) = 1.36 \times 10^{11} \text{ Pa}$ and $p(0) = 3.64 \times 10^{11} \text{ Pa}$.

Solution. The mass within radius r is

$$M(r) = \begin{cases} \frac{4}{3}\pi\rho_c r^3 & 0 \leq r \leq c, \\ \frac{4}{3}\pi [\rho_c c^3 + \rho_m(r^3 - c^3)] & c \leq r \leq b, \end{cases} \quad (7)$$

and so, with $g_0 = GM(r)/r^2$ as in Example 1,

$$g_0(r) = \begin{cases} \frac{4}{3}\pi G\rho_c r & 0 \leq r \leq c, \\ \frac{4}{3}\pi G \left[\rho_m r + (\rho_c - \rho_m)\frac{c^3}{r^2} \right] & c \leq r \leq b. \end{cases} \quad (8)$$

In the mantle, g_0 is the sum of the linearly increasing field of a homogeneous planet of density ρ_m and the inverse-square field of the excess core mass $\frac{4}{3}\pi(\rho_c - \rho_m)c^3$. These

two terms vary in opposite senses with r , which is why g_0 is so nearly uniform through the Earth's mantle. Note that g_0 is continuous at $r = c$, as it must be since $M(r)$ is continuous.

The pressure follows from $dp/dr = -\rho g_0$ with $p(b) = 0$. In the mantle,

$$p(r) = \rho_m \int_r^b g_0(r') dr' = \frac{2}{3}\pi G \rho_m^2 (b^2 - r^2) + \frac{4}{3}\pi G \rho_m (\rho_c - \rho_m) c^3 \left(\frac{1}{r} - \frac{1}{b} \right), \quad c \leq r \leq b. \quad (9)$$

The first term is the pressure in a homogeneous planet of density ρ_m , while the second is the additional pressure needed to support the mantle against the attraction of the excess core mass. Setting $r = c$ gives the pressure at the CMB,

$$p(c) = \frac{2}{3}\pi G \rho_m^2 (b^2 - c^2) + \frac{4}{3}\pi G \rho_m (\rho_c - \rho_m) c^2 \left(1 - \frac{c}{b} \right). \quad (10)$$

Within the core the density is uniform and g_0 is linear in r , so the calculation of Example 1 applies with the constant of integration fixed by continuity of p at the CMB,

$$p(r) = p(c) + \frac{2}{3}\pi G \rho_c^2 (c^2 - r^2), \quad 0 \leq r \leq c. \quad (11)$$

The pressure is continuous at $r = c$, but its gradient $-\rho g_0$ jumps by the factor ρ_c/ρ_m , since g_0 is continuous while ρ is not.

We now put in the numbers. From eq. (7) with $r = b$, the mass is

$$M = \frac{4}{3}\pi \left[\rho_c c^3 + \rho_m (b^3 - c^3) \right] = 6.02 \times 10^{24} \text{ kg}, \quad (12)$$

within one per cent of the Earth's mass. The moment of inertia about any axis through the centre of a spherically symmetric body is $I = \frac{8}{3}\pi \int_0^b \rho r^4 dr$, and hence

$$I = \frac{8}{15}\pi \left[\rho_c c^5 + \rho_m (b^5 - c^5) \right] = 8.47 \times 10^{37} \text{ kg m}^2, \quad \frac{I}{Mb^2} = 0.347. \quad (13)$$

For a homogeneous sphere $I/(Mb^2) = 0.4$, and the observed value of 0.3307 for the Earth is one of the classical indications that its density increases with depth. The two-layer model accounts for most of the reduction, but not all of it, because within each layer the real density increases with depth under compression, which the model neglects. The gravitational acceleration from eq. (8) is $g_0(b) = 9.90 \text{ m s}^{-2}$ at the surface and $g_0(c) = 10.70 \text{ m s}^{-2}$ at the CMB, the values in PREM being 9.82 and 10.68 m s^{-2} . For the pressure, the two terms in eq. (10) evaluate to $8.06 \times 10^{10} \text{ Pa}$ and $4.49 \times 10^{10} \text{ Pa}$, giving

$$p(c) = 1.26 \times 10^{11} \text{ Pa}, \quad (14)$$

and the core then adds $\frac{2}{3}\pi G \rho_c^2 c^2 = 2.05 \times 10^{11} \text{ Pa}$, so that from eq. (11)

$$p(0) = 3.30 \times 10^{11} \text{ Pa}. \quad (15)$$

These are within 8 and 9 per cent of the PREM values of $1.36 \times 10^{11} \text{ Pa}$ and $3.64 \times 10^{11} \text{ Pa}$, and almost twice the central pressure of the homogeneous model of Example 1. The model errs on the low side for the same reason that $I/(Mb^2)$ errs on the high side: neglecting self-compression places too little mass at depth. A model with only two free densities nevertheless reproduces the mass, moment of inertia, and pressures of the Earth to within about ten per cent, which illustrates how strongly the gross structure of a planet is controlled by the existence of a dense core.

Example 3: Centrifugal potential and Y_{20} . For $\mathbf{\Omega} = \Omega \hat{\mathbf{e}}_3$ show that the centrifugal potential $\psi = \frac{1}{2}(\Omega_i \Omega_j - \Omega_k \Omega_k \delta_{ij}) x_i x_j$ defined in the lecture can be written

$$\psi = -\frac{1}{3}\Omega^2 r^2 + \frac{1}{3}\Omega^2 r^2 P_2(\cos \theta),$$

with θ the colatitude and P_2 the Legendre polynomial of degree two, and hence that its aspherical part is proportional to $r^2 Y_{20}$. Explain why the spherical part can be absorbed into the reference model, and deduce that all first-order perturbations of a slowly rotating hydrostatic planet have Y_{20} angular dependence. Using the lecture's definition of the ellipticity, show that the surface of the planet is $r(\theta) = b [1 - \frac{2}{3}\epsilon(b) P_2(\cos \theta)]$ and verify that $\epsilon(b)$ is the difference between the equatorial and polar radii divided by the mean radius.

Solution. With $\Omega_i = \Omega \delta_{i3}$ the definition gives

$$\psi = \frac{1}{2} (\Omega^2 x_3^2 - \Omega^2 x_k x_k) = -\frac{1}{2}\Omega^2 (x_1^2 + x_2^2) = -\frac{1}{2}\Omega^2 r^2 \sin^2 \theta, \quad (16)$$

which is minus one half of the square of the distance from the rotation axis times Ω^2 . Its gradient is $-\Omega^2(x_1, x_2, 0)$, the outward centrifugal acceleration $-\mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{x})$, as it should be. Recalling that $P_2(\cos \theta) = \frac{1}{2}(3 \cos^2 \theta - 1)$, we have

$$\sin^2 \theta = 1 - \cos^2 \theta = 1 - \frac{1}{3} [2P_2(\cos \theta) + 1] = \frac{2}{3} [1 - P_2(\cos \theta)], \quad (17)$$

and substituting into eq. (16) gives

$$\psi = -\frac{1}{3}\Omega^2 r^2 + \frac{1}{3}\Omega^2 r^2 P_2(\cos \theta), \quad (18)$$

as required. The zonal spherical harmonic of degree two is $Y_{20} = \sqrt{5/(4\pi)} P_2(\cos \theta)$, with the normalisation $\int |Y_{20}|^2 dS = 1$ over the unit sphere, and so the aspherical part of ψ is

$$\psi^{\text{asph}} = \sqrt{\frac{4\pi}{5}} \frac{1}{3} \Omega^2 r^2 Y_{20}. \quad (19)$$

As a check, eq. (16) gives $\nabla^2 \psi = -2\Omega^2$, and since $\nabla^2 r^2 = 6$ the spherical part of eq. (18) supplies the whole of this, while $r^2 Y_{20}$ is a solid harmonic and hence contributes nothing.

The spherical part $-\frac{1}{3}\Omega^2 r^2$ depends on r alone. Adding it to the gravitational potential of the reference model simply changes the zeroth-order equilibrium equation to $dp_0/dr + \rho_0(g_0 - \frac{2}{3}\Omega^2 r) = 0$, that is, the spherical model feels a slightly reduced effective gravity. The relative change, $\frac{2}{3}\Omega^2 r/g_0$, is at most $\frac{2}{3}\Omega^2 b^3/(GM) \approx 2 \times 10^{-3}$ for the Earth, so this is a first-order correction to a spherically symmetric model. It changes the reference values of p_0 and g_0 by a fraction of order the small parameter, but does not alter the spherical symmetry of the reference state, and it is this spherical symmetry alone that the first-order theory relies upon. We may therefore regard the spherical part as already included in ρ_0 , p_0 , and ϕ_0 , and take $\psi_1 = \psi^{\text{asph}}$. This is precisely the freedom used in the lecture when the perturbations were assumed to average to zero over spherical surfaces.

Now consider the first-order problem. The boundary value problem for ϕ_1 is linear in ϕ_1 and ψ_1 , its coefficients $4\pi G\rho'_0/g_0$ and $4\pi G\rho_0/g_0$ depend on r alone, and the Laplacian commutes with rotations. Substituting the separated form $\phi_1 = f(r)Y_{20}$, the angular dependence factors out of every term, leaving an ordinary differential equation in r for f whose forcing is proportional to r^2 . Any other spherical harmonic component of ϕ_1 would satisfy the same problem with no forcing, and must vanish for the perturbation to be unique.¹ Hence ϕ_1 , and therefore $\gamma_1 = \phi_1 + \psi_1$, is proportional to Y_{20} . The remaining first-order fields are given in terms of γ_1 by the algebraic relations $p_1 = -\rho_0\gamma_1$, $h_1 = -\gamma_1/g_0$, and $\rho_1 = (\rho'_0/g_0)\gamma_1$, with coefficients depending on r alone, and so all have the same Y_{20} dependence. Since Y_{20} is independent of longitude and even in $\cos\theta$, the perturbed planet is axisymmetric and symmetric about its equatorial plane.

Finally, the lecture defines the ellipticity by $\gamma_1 = \sqrt{4\pi/5} \frac{2}{3} r g_0 \epsilon Y_{20} = \frac{2}{3} r g_0 \epsilon P_2(\cos\theta)$, and therefore the surface displacement is

$$h_1 = -\frac{\gamma_1}{g_0} \Big|_{r=b} = -\frac{2}{3} b \epsilon(b) P_2(\cos\theta), \quad r(\theta) = b \left[1 - \frac{2}{3} \epsilon(b) P_2(\cos\theta) \right]. \quad (20)$$

At the equator $P_2(0) = -\frac{1}{2}$ and at the poles $P_2(\pm 1) = 1$, so the equatorial and polar radii are

$$r_{\text{eq}} = b \left[1 + \frac{1}{3} \epsilon(b) \right], \quad r_{\text{pol}} = b \left[1 - \frac{2}{3} \epsilon(b) \right], \quad (21)$$

and their difference is $r_{\text{eq}} - r_{\text{pol}} = b \epsilon(b)$. For $\epsilon > 0$ the planet is oblate, with the equatorial bulge twice as far from the mean sphere as the polar depression, as required by the vanishing of the spherical average of P_2 . That average also shows that b is the mean radius of the perturbed surface, and that its volume is unchanged to first order. Hence $\epsilon(b) = (r_{\text{eq}} - r_{\text{pol}})/b$, which is the geometrical interpretation quoted in the lecture; for the Earth $r_{\text{eq}} - r_{\text{pol}} \approx 21.4 \text{ km}$ and $\epsilon(b) \approx 1/298$. The lesson is that the entire aspherical structure of a slowly rotating hydrostatic planet is described by one radial function, because the forcing has a single angular component and the problem is linear.

Example 4: Clairaut's equation for a homogeneous planet. Show that for a planet of uniform density ρ_0 one has $8\pi G\rho_0/g_0 = 6/r$, so that Clairaut's equation reduces to $\epsilon'' + 6\epsilon'/r = 0$, whose regular solution is $\epsilon = \text{const}$. Apply the surface boundary condition to obtain Newton's result

$$\epsilon = \frac{5}{4} \frac{\Omega^2 b^3}{GM} = \frac{5}{4} m,$$

where $m = \Omega^2 b^3/(GM)$. Evaluate $1/\epsilon$ for a homogeneous Earth, compare with the observed value of 298, and explain the sign of the discrepancy.

Solution. For constant ρ_0 we found in Example 1 that $g_0 = \frac{4}{3}\pi G\rho_0 r$, and so

$$\frac{8\pi G\rho_0}{g_0} = \frac{8\pi G\rho_0}{\frac{4}{3}\pi G\rho_0 r} = \frac{6}{r}. \quad (22)$$

¹ The one exception is the degree-one part, which corresponds to a rigid translation of the whole planet and is removed by keeping the centre of mass at the origin.

More generally, $g_0 = \frac{4}{3}\pi G \bar{\rho}(r) r$ with $\bar{\rho}(r)$ the mean density inside radius r , and the coefficient in Clairaut's equation is $6\rho_0(r)/[r\bar{\rho}(r)]$; it is the ratio of the local density to the mean density within r that controls the solution. Substituting eq. (22) into Clairaut's equation,

$$\frac{d^2\epsilon}{dr^2} + \frac{6}{r} \left(\frac{d\epsilon}{dr} + \frac{\epsilon}{r} \right) - \frac{6\epsilon}{r^2} = \frac{d^2\epsilon}{dr^2} + \frac{6}{r} \frac{d\epsilon}{dr} = 0, \quad (23)$$

the terms in ϵ/r^2 cancelling exactly. This is an equation of Euler type, and trying $\epsilon = r^n$ gives $n(n-1) + 6n = n(n+5) = 0$, so the general solution is

$$\epsilon = A + Br^{-5}. \quad (24)$$

The second solution is singular at the centre and violates the condition $\epsilon'(0) = 0$, so we must take $B = 0$: within a homogeneous planet the ellipticity is the same on every level surface, and all the level surfaces are similar spheroids.

The constant A is fixed by the surface condition. With $\epsilon'(b) = 0$ the boundary condition from the lecture,

$$\left. \frac{d\epsilon}{dr} \right|_{r=b} = \frac{1}{b} \left[\frac{5\Omega^2 b^3}{2GM} - 2\epsilon(b) \right], \quad (25)$$

becomes $2\epsilon = 5\Omega^2 b^3 / (2GM)$, and so

$$\epsilon = \frac{5}{4} \frac{\Omega^2 b^3}{GM} = \frac{5}{4} m, \quad m = \frac{\Omega^2 b^3}{GM} = \frac{\Omega^2 b}{g(b)} = \frac{3\Omega^2}{4\pi G \rho_0}, \quad (26)$$

which is Newton's result. The parameter m is the ratio of the centrifugal to the gravitational acceleration at the equator; the last form shows it to be, up to a numerical factor, the ratio of centrifugal to gravitational potentials used in the lecture to define slow rotation. For the Earth, $m = (7.292 \times 10^{-5})^2 (6.371 \times 10^6)^3 / (3.986 \times 10^{14}) = 3.45 \times 10^{-3}$, so that

$$\epsilon = \frac{5}{4} m = 4.31 \times 10^{-3}, \quad \frac{1}{\epsilon} = 232, \quad (27)$$

compared with the observed $1/\epsilon = 298$. The homogeneous planet is thus too flattened by about 30 per cent.²

To understand the sign of the discrepancy it helps to see where the factor $5/4$ comes from. Write $\gamma_1 = \frac{2}{3} r g_0 \epsilon P_2(\cos \theta)$ as in Example 3. Displacing the surface of a homogeneous planet by $h_1 = -\frac{2}{3} b \epsilon P_2$ is equivalent, to first order, to placing a surface mass density $\sigma = \rho_0 h_1$ on the sphere $r = b$. The interior potential of a surface density $\sigma_2 P_2(\cos \theta)$ on a sphere of radius b is $-\frac{4}{5} \pi G b \sigma_2 (r/b)^2 P_2(\cos \theta)$, a standard consequence of the Legendre expansion of $1/\|\mathbf{x} - \mathbf{x}'\|$, and so

$$\phi_1 = \frac{8}{15} \pi G \rho_0 \epsilon r^2 P_2(\cos \theta), \quad r < b. \quad (28)$$

² Newton obtained $1/230$ in the "Principia" using a slightly different value of m . The French geodetic expeditions to Lapland and Peru in the 1730s confirmed that the Earth is oblate, against the prolate Earth favoured by the Cassinis, but accurate values of the flattening came only later from geodesy and from the analysis of the Moon's motion. Clairaut's theory of 1743 explained why the Earth is less flattened than a homogeneous planet.

This is positive at the poles and negative at the equator: the extra mass in the equatorial bulge deepens the potential well there. Adding the aspherical centrifugal potential of eq. (18) and equating to $\gamma_1 = \frac{2}{3}rg_0\epsilon P_2 = \frac{8}{9}\pi G\rho_0\epsilon r^2 P_2$ gives

$$\frac{8}{9}\pi G\rho_0\epsilon = \frac{8}{15}\pi G\rho_0\epsilon + \frac{1}{3}\Omega^2 \implies \epsilon = \frac{15\Omega^2}{16\pi G\rho_0} = \frac{5}{4}m, \quad (29)$$

in agreement with eq. (26), and providing an independent check on the signs in Clairaut's equation and its surface condition. The proportionality to r^2 of every term is the reason ϵ is constant. Had we ignored the self-gravity of the bulge, dropping the first term on the right of eq. (29), we would have found $\epsilon = 3\Omega^2/(8\pi G\rho_0) = \frac{1}{2}m$, the value for a massless fluid envelope surrounding a point mass, for which $1/\epsilon = 580$ for the Earth. The self-attraction of the equatorial bulge therefore amplifies the flattening of a homogeneous planet by a factor of 5/2.

The real Earth lies between these two limits. In a centrally condensed planet the equatorial bulge contains a smaller fraction of the total mass, its self-attraction is weaker, and the flattening is correspondingly smaller than $\frac{5}{4}m$, though larger than $\frac{1}{2}m$. The observed $1/\epsilon = 298 > 232$ is therefore a further indication, independent of the moment of inertia, that the Earth's density increases with depth. Solving Clairaut's equation by shooting for the two-layer model of Example 2 gives $1/\epsilon(b) = 285$ at the surface and $1/\epsilon = 374$ at the CMB, so that ϵ increases outwards from the centre. The same trend is seen in the lower panel of the figure in Lecture 21, where $1/\epsilon$ is about 300 at the surface and increases inwards, and it is a general property of solutions of Clairaut's equation for a density that does not increase outwards. The level surfaces deep within a planet are less flattened than its outer surface because they feel the centrifugal potential, which grows as r^2 , less strongly relative to the gravitational attraction of the mass within them.

Example 5: Hydrostatic level surfaces. In the lecture it was shown that in a hydrostatic planet the level surfaces of p , ρ , and γ coincide. Verify this directly for the first-order solution of the slowly rotating problem, $p_1 = -\rho_0\gamma_1$ and $\rho_1 = (\rho'_0/g_0)\gamma_1$, by showing that the level surfaces of all three fields through the reference radius r are displaced radially by the same amount $-\gamma_1/g_0$. Show also that the radial component of the first-order equilibrium equation, $\nabla p_1 + \rho_0\nabla\gamma_1 + \rho_1g_0\hat{\mathbf{r}} = \mathbf{0}$, is then satisfied identically, and interpret the function $\epsilon(r)$ in the light of this result.

Solution. Consider a field $F = F_0(r) + sF_1(r, \theta, \varphi) + \dots$ whose zeroth-order part depends on r alone. The level surface of F that passes through radius r in the reference model is, to first order, $r + s\delta r_F(\theta, \varphi)$, where

$$F_0(r + s\delta r_F) + sF_1(r, \theta, \varphi) = F_0(r) \implies \delta r_F = -\frac{F_1}{F'_0}, \quad (30)$$

provided $F'_0 \neq 0$. This is the same expansion that produced the boundary condition $p_1 + p'_0h_1 = 0$ on the perturbed surface in the lecture, the outer surface being the level surface $p = 0$. Applying eq. (30) in turn to the three fields, and using the zeroth-order

hydrostatic equation $p'_0 = -\rho_0 g_0$ together with $\gamma'_0 = \phi'_0 = g_0$, we find

$$\delta r_p = -\frac{p_1}{p'_0} = -\frac{-\rho_0 \gamma_1}{-\rho_0 g_0} = -\frac{\gamma_1}{g_0}, \quad (31)$$

$$\delta r_\rho = -\frac{\rho_1}{\rho'_0} = -\frac{1}{\rho'_0} \frac{\rho'_0}{g_0} \gamma_1 = -\frac{\gamma_1}{g_0}, \quad (32)$$

$$\delta r_\gamma = -\frac{\gamma_1}{\gamma'_0} = -\frac{\gamma_1}{g_0}. \quad (33)$$

The three displacements are identical, and equal to the surface displacement h_1 of the lecture evaluated at radius r . Thus at every depth the perturbed level surfaces of pressure, density, and gravity potential coincide, as the exact theory requires. Where $\rho'_0 = 0$ the density has no level surfaces to speak of, but there $\rho_1 = 0$ as well, and there is nothing to check.

The relations for p_1 and ρ_1 can be read in the same way. Writing $h(r, \theta, \varphi) = -\gamma_1/g_0$ for the displacement of the level surface through r , we have

$$p_1 = -\rho_0 \gamma_1 = -p'_0 h, \quad \rho_1 = \frac{\rho'_0}{g_0} \gamma_1 = -\rho'_0 h, \quad (34)$$

which are exactly the changes at a fixed point produced by displacing the spherical profiles $p_0(r)$ and $\rho_0(r)$ radially outwards through h . The first-order solution is, in other words, nothing more than the reference model with each of its level surfaces carried onto the corresponding level surface of the perturbed gravity potential.

Now take the radial component of the first-order equilibrium equation. Substituting $p_1 = -\rho_0 \gamma_1$,

$$\frac{\partial p_1}{\partial r} + \rho_0 \frac{\partial \gamma_1}{\partial r} + \rho_1 g_0 = -\rho'_0 \gamma_1 - \rho_0 \frac{\partial \gamma_1}{\partial r} + \rho_0 \frac{\partial \gamma_1}{\partial r} + \rho_1 g_0 = g_0 \left(\rho_1 - \frac{\rho'_0}{g_0} \gamma_1 \right), \quad (35)$$

which vanishes by the expression for ρ_1 . The tangential components are satisfied because $p_1 + \rho_0 \gamma_1 = 0$ identically. Note that the argument can be run in the other direction: once the tangential components have given $p_1 = -\rho_0 \gamma_1$, the radial component yields $\rho_1 = (\rho'_0/g_0) \gamma_1$ directly, without taking the curl of the equation as was done in the lecture. The two routes must agree, because the curl of the equilibrium equation is a consequence of the equation itself.

Finally, with $\gamma_1 = \frac{2}{3} r g_0 \epsilon(r) P_2(\cos \theta)$ the common displacement is

$$h = -\frac{\gamma_1}{g_0} = -\frac{2}{3} r \epsilon(r) P_2(\cos \theta), \quad (36)$$

which is eq. (20) of Example 3 with b replaced by r . Every level surface of the perturbed planet is therefore an oblate spheroid of mean radius r whose equatorial and polar radii differ by $r \epsilon(r)$, and $\epsilon(r)$ is the ellipticity of the level surface of mean radius r . The lecture's geometrical interpretation of $\epsilon(b)$ is the special case of the outer surface, and the same interpretation applies to the CMB and to any other internal boundary, each of which must be a level surface in a hydrostatic planet. This is why the profile of $1/\epsilon$ plotted in the lecture, and computed for the two-layer model in Example 4, can be read directly as the flattening of the Earth's internal structure: a two-layer Earth in hydrostatic equilibrium would have a CMB with equatorial and polar radii differing by $c/374 \approx 9$ km.